

SQL DATA ANALYTICS PORTFOLIO

Urban Mobility IQ

Ride-Sharing Data Analysis Using SQL

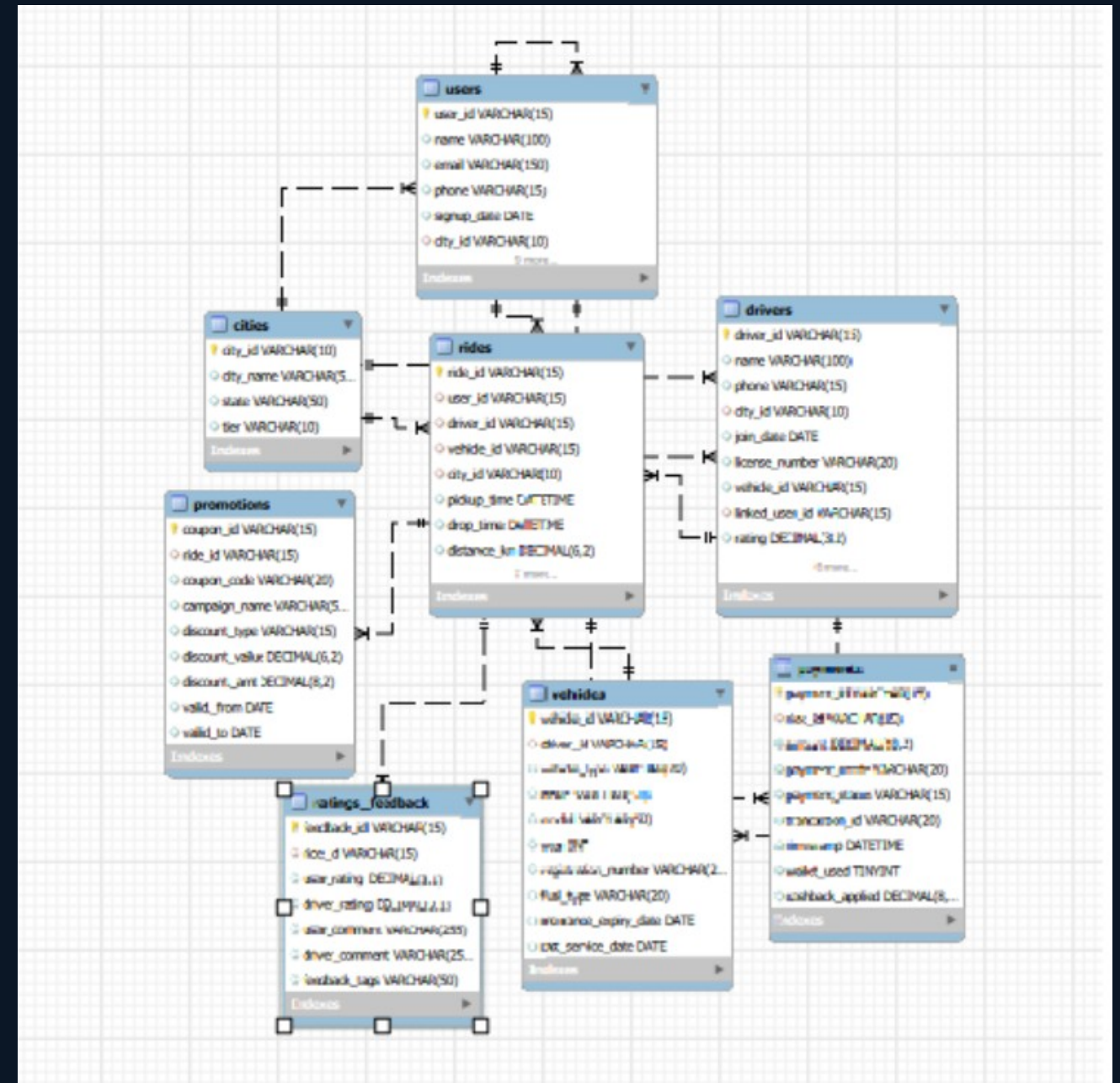
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01 Introduction

A relational view of demand, performance and platform economics

This project analyses ride demand, customer behaviour, driver performance, revenue patterns, payment activity, promotions, ratings and operations using SQL.

JOINS · AGGREGATIONS · GROUP BY · CASE WHEN
SUBQUERIES · CTEs · WINDOW FUNCTIONS · RANK() · DENSE_RANK()



02

Objective

Six questions that frame the analysis

01

Analyse ride demand across cities

02

Identify revenue-driving markets

03

Understand vehicle-type preferences

04

Evaluate driver and customer performance

05

Analyse cancellations and payment behaviour

06

Evaluate promotions, ratings and operational trends

03 Expected Insights

01

Identify high-demand cities

02

Understand vehicle preferences

03

Analyse revenue and payment patterns

04

Identify cancellation drivers

05

Evaluate top-performing drivers and customers

04

Dataset Overview

Eight relational tables · 126,284 total records

▀ Users

3,000 records

▀ Vehicles

800 records

▀ Rides

45,000 records

▀ Promotions

6,802 records

▀ Drivers

800 records

▀ Cities

10 records

▀ Payments

40,184 records

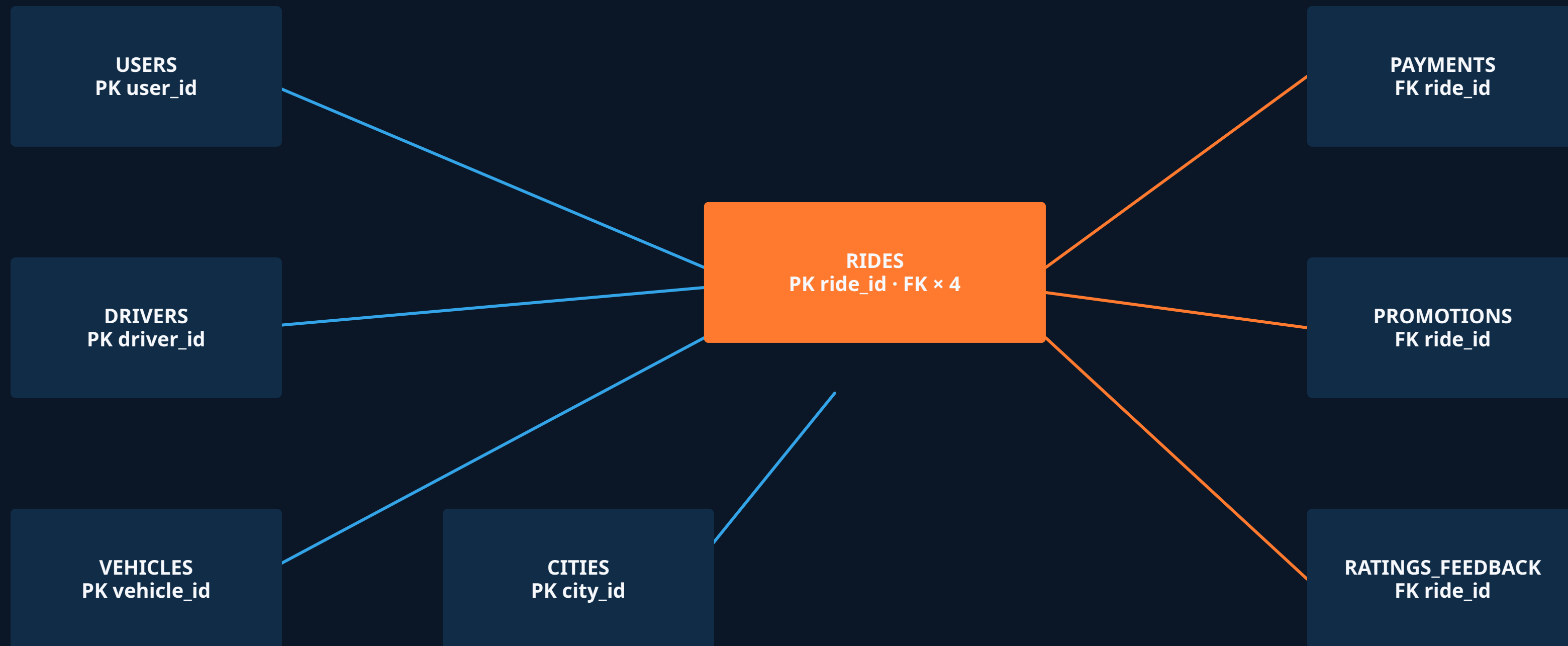
▀ Ratings_Feedback

29,688 records

05

Database Structure

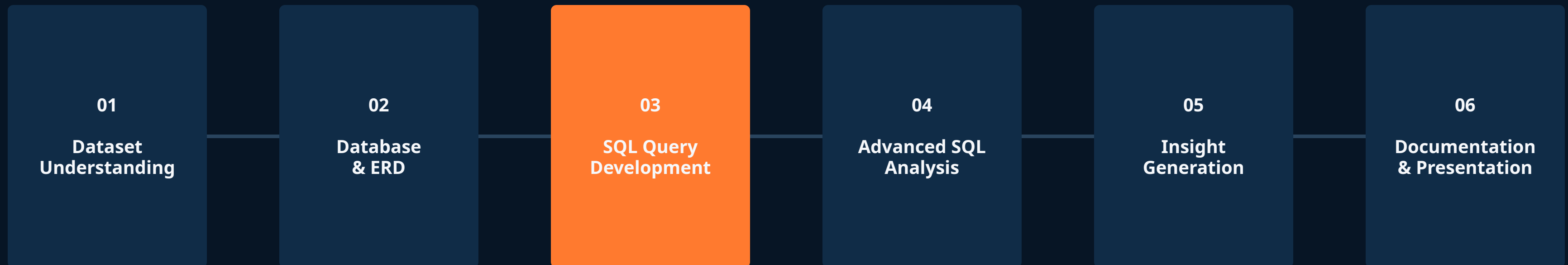
PK primary key · FK foreign key



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Project Map

From raw tables to an executive-ready narrative



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Data Understanding & Preparation

✓ Understand table structures

✓ Identify primary / foreign keys

✓ Check missing values

✓ Validate data types

✓ Check duplicates

✓ Standardise values

✓ Verify relationships

✓ Validate business logic

08

Questions to Answer

Twelve business questions across demand, revenue, operations and experience

01 How many rides were completed?

02 Which cities have the highest ride volume?

03 Which cities generate the most revenue?

04 Which vehicle types are most popular?

05 Which vehicle types generate higher fares?

06 What are the major cancellation reasons?

07 Which drivers generate the highest revenue?

08 Which customers spend the most?

09 What payment methods are most used?

10 How effective are promotions?

11 What is the payment success / failure pattern?

12 How do ratings vary across the platform?

09 Ride Status

How many rides were completed? What is the ride status distribution?

```
SELECT ride_status,  
       COUNT(*) AS ride_count,  
       SUM(CASE WHEN ride_status = 'Completed'  
               THEN 1 ELSE 0 END) AS completed  
FROM rides  
GROUP BY ride_status  
ORDER BY ride_count DESC;
```

Result Grid   Filter Rows:			
	ride_status	ride_count	completed
▶	completed	38101	38101
	cancelled	4604	0
	no_show	2295	0

BUSINESS INSIGHT — Completed rides constitute the majority with 38,101 out of 45,000 rides, highlighting strong platform engagement.

10 City Performance

Which cities have the highest ride volume?
What is the revenue distribution by city?

```
SELECT c.city_name,  
       COUNT(r.ride_id) AS rides,  
       SUM(r.total_fare) AS revenue  
FROM rides r  
JOIN cities c ON c.city_id = r.city_id  
GROUP BY c.city_name  
ORDER BY revenue DESC;
```

Result Grid   Filter Rows:			
	city_name	rides	revenue
▶	Hyderabad	5723	458414.69
	Mumbai	5284	436666.28
	Pune	4706	425743.40
	Kolkata	4950	419466.08
	Delhi	4300	385735.89
	Lucknow	4331	362209.55
	Ahmedabad	3894	351473.08
	Bengaluru	4080	345017.99
	Jaipur	3749	341607.57
	Chennai	3983	335483.67

BUSINESS INSIGHT — Hyderabad leads with 5,723 rides and highest revenue, followed by Mumbai and Kolkata, indicating key target markets.

11 Vehicle & Ride Analysis

What are the popular vehicle types by ride count?

What are the average fares by vehicle type?

```
SELECT v.vehicle_type,  
COUNT(r.ride_id) AS rides,  
ROUND(AVG(r.total_fare), 2) AS avg_fare  
FROM rides r  
JOIN vehicles v ON v.vehicle_id=r.vehicle_id  
GROUP BY v.vehicle_type  
ORDER BY rides DESC;
```

Result Grid   Filter Rows:			
	vehicle_type	rides	avg_fare
▶	Mini	16661	92.86
	Bike	10835	46.92
	Auto	8793	67.74
	Prime	6106	129.01
	XL	2605	162.37

BUSINESS INSIGHT — Mini vehicles dominate ride volume at 16,661 rides, while XL and Prime command higher average fares of ₹192.34 and ₹151.54 respectively.

12 Driver & Customer Analysis

Who are the top drivers by revenue?

Who are the top customers by spending?

```
SELECT d.name, SUM(r.total_fare) revenue
FROM drivers d JOIN rides r USING(driver_id)
GROUP BY d.driver_id, d.name
ORDER BY revenue DESC LIMIT 10;
```

```
SELECT user_id, SUM(total_fare) spend
FROM rides GROUP BY user_id
ORDER BY spend DESC LIMIT 10;
```

	name	revenue
▶	Dipta Kannan	13563.43
	Yahvi Kapadia	13268.33
	Orinder Dhaliwal	12733.52
	Faqid Lalla	12659.11
	Fitan Chana	12547.37
	Meghana Kohli	12489.64
	Yashawini Parikh	12124.96
	Naveen Ahuja	12116.50

	user_id	spend
▶	USR002542	3224.01
	USR001062	3089.75
	USR000579	3073.49
	USR000591	3022.53
	USR000137	2987.61
	USR002999	2921.78
	USR002534	2876.25
	USR002317	2840.23
	USR001392	2825.88
	USR001334	2802.82

BUSINESS INSIGHT — Top driver Dipta Kannan earned ₹13,547.66, and top customer USR002542 spent ₹3,224.01, reflecting the platform's key contributors.

13 Cancellation, Payment & Promotion

Cancellation reasons · payment modes and status ·
best-performing campaign

```
SELECT cancellation_reason, COUNT(*) total
FROM rides WHERE ride_status='Cancelled'
GROUP BY cancellation_reason;
```

```
SELECT payment_status, payment_method, COUNT(*)
FROM payments GROUP BY payment_status, payment_method;
```

```
SELECT campaign_name, SUM(discount_amt) discount
FROM promotions GROUP BY campaign_name;
```

	cancellation_reason	total
▶	Driver cancelled	750
	Driver not reachable	752
	User cancelled	781
	Long wait time	779
	Change of plans	780
	Vehicle breakdown	762

	payment_status	payment_mode	COUNT(*)
▶	Success	Wallet	5403
	Success	Cash	5250
	Success	UPI	5256
	Refunded	UPI	1228
	Failed	UPI	1522
	Success	Credit Card	5303
	Refunded	Wallet	1284
	Success	Debit Card	5151
	Failed	Credit Card	1474

	campaign_name	discount
▶	First ride discount	23604.30
	New user welcome offer	41358.83
	Loyalty reward	32546.24
	Festive season offer	52045.23
	Weekend ride offer	17363.66
	Monsoon special	11762.45

BUSINESS INSIGHT — Payment success is strong at 26,363 transactions, with the Festive Season Offer driving ₹52,045.23 in discounts.

14 Advanced SQL Analysis

Who are the top-revenue drivers in each city based on completed rides?

```
WITH driver_revenue AS (  
  SELECT city_id, driver_id,  
         SUM(total_fare) AS revenue  
  FROM rides  
 WHERE ride_status='Completed'  
 GROUP BY city_id, driver_id  
)  
SELECT *, RANK() OVER(  
  PARTITION BY city_id ORDER BY revenue DESC  
) AS city_rank  
FROM driver_revenue;
```

Result Grid		Filter Rows:		
	city_id	driver_id	revenue	city_rank
▶	AMD	DRV00252	13230.98	1
	AMD	DRV00318	12097.61	2
	AMD	DRV00536	10136.98	3
	AMD	DRV00287	10127.40	4
	AMD	DRV00138	10078.16	5
	AMD	DRV00446	9743.53	6
	AMD	DRV00327	9719.92	7
	AMD	DRV00706	9644.02	8
	AMD	DRV00245	9639.66	9

BUSINESS INSIGHT — Advanced SQL converts transactional rows into reusable analytical layers and local performance benchmarks.

15 Window Function Analysis

Identify top-performing drivers within each city using rankings.

```
WITH x AS (  
  SELECT c.city_name, d.name,  
         SUM(r.total_fare) revenue  
  FROM rides r JOIN cities c USING(city_id)  
  JOIN drivers d USING(driver_id)  
  GROUP BY c.city_name, d.name  
)  
SELECT *,  
       RANK() OVER(PARTITION BY city_name ORDER BY revenue DESC) rnk,  
       DENSE_RANK() OVER(PARTITION BY city_name ORDER BY revenue DESC)  
       dense_rnk  
FROM x;
```

Result Grid  Filter Rows: Export:  Wrap					
	city_name	name	revenue	rnk	dense_rnk
▶	Ahmedabad	Yahvi Kapadia	13268.33	1	1
	Ahmedabad	Naveen Ahuja	12116.50	2	2
	Ahmedabad	Laksh Babu	10142.04	3	3
	Ahmedabad	Jason Pillai	10137.39	4	4
	Ahmedabad	Watika Bal	10132.48	5	5
	Ahmedabad	Orinder Bhalla	9821.41	6	6
	Ahmedabad	Yashodhara Mane	9743.06	7	7
	Ahmedabad	Ria Thakur	9698.65	8	8
	Ahmedabad	Ishita Nagarajan	9663.77	9	9
	Ahmedabad	Sanya Majumdar	9052.69	10	10

BUSINESS INSIGHT — Ranking drivers by city reveals local leaderboards, enabling focused performance incentives.

16 CTE & Subquery Analysis

Drivers above a ride threshold · performance vs city average · customer spend ranks · filtered aggregates

```
WITH driver_totals AS (  
  SELECT driver_id, city_id, COUNT(*) rides,  
         SUM(total_fare) revenue  
  FROM rides GROUP BY driver_id, city_id  
) , city_avg AS (  
  SELECT city_id, AVG(revenue) avg_revenue  
  FROM driver_totals GROUP BY city_id  
)  
SELECT d.* FROM driver_totals d  
JOIN city_avg c USING(city_id)  
WHERE d.rides > 50 AND d.revenue > c.avg_revenue;
```

Result Grid		Filter Rows:		
	driver_id	city_id	rides	revenue
▶	DRV00080	HYD	73	7097.26
	DRV00027	MUM	65	10661.69
	DRV00200	KOL	71	6648.09
	DRV00147	MUM	61	5904.38
	DRV00319	PUNE	77	7144.18
	DRV00602	LKO	61	6031.45
	DRV00203	CHN	58	7541.03
	DRV00186	KOL	73	6705.82
	DRV00707	MUM	61	5862.79
	DRV00800	LKO	78	5791.40

BUSINESS INSIGHT — CTEs and subqueries enable modular, efficient analysis of complex business questions.

17

Key Findings

Six conclusions supported by the SQL analysis

01

Hyderabad leads platform ride demand.

02

Mini vehicles dominate ride volume.

03

XL and Prime generate significantly higher average fares.

04

87.3% of users are active.

05

Higher surge multipliers correspond with higher average fares.

06

Cancellation behaviour presents operational improvement opportunities.

18 Conclusion & Learning Outcomes

- 01 Strengthened SQL querying skills
- 02 Applied joins and aggregations to real-world data
- 03 Used CTEs for structured analysis
- 04 Applied window functions for ranking
- 05 Analysed customer and driver behaviour
- 06 Converted query results into business insights
- 07 Improved relational database understanding

SQL SKILLS DEMONSTRATED

SELECT
WHERE
GROUP BY · ORDER BY
JOIN · CASE WHEN
SUBQUERIES · CTEs
WINDOW FUNCTIONS
RANK · DENSE_RANK
AGGREGATE FUNCTIONS

THANK YOU

From SQL Queries to Business Decisions

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QUESTIONS?

SQL | MySQL | Excel | Python | Data Analytics